

Limits & Derivatives

ANSWER KEY



Evaluating limits by direct substitution

- 1 Evaluate $\lim_{x \rightarrow 2} (3x^2 - 4x + 1)$.
 $3(4) - 8 + 1 = 12 - 8 + 1 = 5$.
 - 2 Evaluate $\lim_{x \rightarrow -1} (x^3 + 2x - 5)$.
 $(-1)^3 + 2(-1) - 5 = -1 - 2 - 5 = -8$.
 - 3 Evaluate $\lim_{x \rightarrow 0} \frac{5x + 7}{7x + 3}$.
 $\frac{5(0) + 7}{0 + 3} = \frac{7}{3}$.
 - 4 Evaluate $\lim_{x \rightarrow 3} \sqrt{x + 13}$.
 $\sqrt{3 + 13} = \sqrt{16} = 4$.
-

Evaluating limits by factoring (the 0/0 form)

- 5 Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{(x - 3)(x + 3)}$.
 $\frac{x^2 - 9}{x - 3} = x + 3 \rightarrow 6$.
 - 6 Evaluate $\lim_{x \rightarrow -3} \frac{x^2 + 5x + 6}{(x + 2)(x + 3)}$.
 $\frac{x^2 + 5x + 6}{x + 2} = x + 3 \rightarrow 1$.
 - 7 Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 16}{(x - 4)(x + 4)}$.
 $\frac{x^2 - 16}{x - 4} = x + 4 \rightarrow 8$.
 - 8 Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{(x - 1)(x + 1)}$.
 $\frac{x^2 - 1}{(x - 1)(x + 1)} = \frac{x + 1}{x + 2} \rightarrow \frac{2}{-1} = -2$.
-

Limits at infinity

- 9 Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{x^2 - 5}$.
The degrees match, so the limit is the ratio of leading coefficients: 3.
 - 10 Evaluate $\lim_{x \rightarrow \infty} \frac{2x + 7}{x^2 + 1}$.
The denominator has higher degree, so the limit is 0.
 - 11 Evaluate $\lim_{x \rightarrow \infty} \frac{5x^3 - 2}{x^3 + 4x}$.
The degrees match, so the limit is the ratio of leading coefficients: 5.
-

One-sided limits

- 12 A function is defined by $f(x) = x + 1$ for $x < 2$, and $f(x) = 2x - 1$ for $x \geq 2$. Find $\lim_{x \rightarrow 2^-} f(x)$.
Using the first rule: $\lim_{x \rightarrow 2^-} f(x) = 2 + 1 = 3$.
- 13 Using the same piecewise function, find $\lim_{x \rightarrow 2^+} f(x)$.
Using the second rule: $\lim_{x \rightarrow 2^+} f(x) = 2(2) - 1 = 3$.
- 14 Using the same piecewise function, does $\lim_{x \rightarrow 2} f(x)$ exist? State its value.
Yes — both one-sided limits equal 3, so $\lim_{x \rightarrow 2} f(x) = 3$.
-

The power rule

- 15 Differentiate $f(x) = x^5$.
 $f'(x) = 5x^4$.
- 16 Differentiate $f(x) = 3x^4 - 2x^2 + 7$.
 $f'(x) = 12x^3 - 4x$.
- 17 Differentiate $f(x) = \sqrt{x}$.
 $f(x) = x^{1/2}$, so $f'(x) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$.
- 18 Differentiate $f(x) = \frac{1}{x}$.
 $f(x) = x^{-2}$, so $f'(x) = -2x^{-3} = -\frac{2}{x^3}$.
-

The product rule

- 19 Differentiate $f(x) = x^2(3x + 1)$ using the product rule.
 $f'(x) = (2x)(3x + 1) + x^2(3) = 6x^2 + 2x + 3x^2 = 9x^2 + 2x$.
- 20 Differentiate $f(x) = (x + 2)(x - 5)$ using the product rule.
 $f'(x) = (1)(x - 5) + (x + 2)(1) = x - 5 + x + 2 = 2x - 3$.
- 21 Differentiate $f(x) = (2x - 1)(x^2 + 3)$ using the product rule.
 $f'(x) = (2)(x^2 + 3) + (2x - 1)(2x) = 2x^2 + 6 + 4x^2 - 2x = 6x^2 - 2x + 6$.
-

The chain rule

- 22 Differentiate $f(x) = (3x + 1)^4$.
 $f'(x) = 4(3x + 1)^3(3) = 12(3x + 1)^3$.

- 23 Differentiate $f(x) = (x^2 - 5)^3$.
 $f'(x) = 3(x^2 - 5)^2(2x) = 6x(x^2 - 5)^2$.
- 24 Differentiate $f(x) = \sqrt{4x + 9}$.
 $f(x) = (4x + 9)^{1/2}$, so $f'(x) = \frac{1}{2}(4x + 9)^{-1/2}(4) = \frac{2}{\sqrt{4x + 9}}$.
- 25 Differentiate $f(x) = (5 - 2x)^6$.
 $f'(x) = 6(5 - 2x)^5(-2) = -12(5 - 2x)^5$.
-

Derivatives of trigonometric and exponential functions

- 26 Differentiate $f(x) = \sin x + \cos x$.
 $f'(x) = \cos x - \sin x$.
- 27 Differentiate $f(x) = 3\sin x$.
 $f'(x) = 3\cos x$.
- 28 Differentiate $f(x) = e^x + x^2$.
 $f'(x) = e^x + 2x$.
-

Tangent lines

- 29 Find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.
 $f'(x) = 2x$, so $f'(3) = 6$.
- 30 Find the equation of the tangent line to $f(x) = x^2$ at $x = 2$.
 $f(2) = 4$, $f'(2) = 4$. $y - 4 = 4(x - 2) \Rightarrow y = 4x - 4$.
- 31 The graph of $f(x) = x^2 - 2x$ is shown. Find the equation of the tangent line at $x = 3$ (the marked point).
 $f(3) = 9 - 6 = 3$. $f'(x) = 2x - 2$, so $f'(3) = 4$. $y - 3 = 4(x - 3) \Rightarrow y = 4x - 9$.
- 32 Find the x -coordinate(s) where the tangent to $f(x) = x^3 - 3x$ is horizontal.
 $f'(x) = 3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$.
-

Increasing, decreasing, and the second derivative

- 33 For $f(x) = x^2 - 4x + 1$, find $f'(x)$ and determine the interval where f is increasing.
 $f'(x) = 2x - 4$. Increasing when $f'(x) > 0$, i.e. $x > 2$.
- 34 For $f(x) = -x^2 + 6x$, determine the interval where f is decreasing.
 $f'(x) = -2x + 6$. Decreasing when $f'(x) < 0$, i.e. $-2x + 6 < 0 \Rightarrow x > 3$.

- 35 Find the second derivative of $f(x) = x^4 - 2x^2$.
 $f'(x) = 4x^3 - 4x$, so $f''(x) = 12x^2 - 4$.
-

Word problems: rates of change

- 36 This question has two parts. The height of a ball thrown upward is given by $h(t) = -5t^2 + 20t + 2$ (in meters, t in seconds).
- a Find the velocity function $h'(t)$. $h'(t) = -10t + 20$.
 - b Find the velocity at $t = 1$ second, and state whether the ball is rising or falling.
 $h'(1) = -10(1) + 20 = 10$ m/s, which is positive, so the ball is rising.
- 37 This question has two parts. A company's cost to produce x items is $C(x) = 0.01x^2 + 5x + 200$ (in dollars).
- a Find the marginal cost function $C'(x)$. $C'(x) = 0.02x + 5$.
 - b Find the marginal cost at $x = 100$ items, and interpret its meaning.
 $C'(100) = 0.02(100) + 5 = 2 + 5 = 7$ dollars — approximately the cost of producing one more item beyond the 100th.
-

MCQ — Evaluating limits by direct substitution

- 38 Evaluate $\lim_{x \rightarrow 3} (2x^2 - 5x + 3)$.

- a. 12
- b. -6
- c. 18
- d. 6 ✓

$$2(9) - 5(3) + 3 = 18 - 15 + 3 = 6.$$

39 Evaluate $\lim_{x \rightarrow -2} (x^3 - 3x + 4)$.

a. 10

b. -2

c. 2 ✓

d. -10

$$(-2)^3 - 3(-2) + 4 = -8 + 6 + 4 = 2.$$

40 Evaluate $\lim_{x \rightarrow 1} \frac{4x + 9}{x + 5}$.

a. $\frac{9}{5}$

b. $\frac{13}{6}$ ✓

c. $\frac{13}{5}$

d. $\frac{4}{5}$

$$\frac{4(1) + 9}{1 + 5} = \frac{13}{6}.$$

41 Evaluate $\lim_{x \rightarrow 6} \sqrt{x + 10}$.

a. 4 ✓

b. 8

c. $\sqrt{16}$

d. 16

$$\sqrt{6 + 10} = \sqrt{16} = 4.$$

42 Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 3x}{2x + 1}$.

a. $\frac{4}{7}$

b. $\frac{16}{9}$

c. $\frac{4}{8}$

d. $\frac{4}{9}$ ✓

$$\frac{16 - 12}{9} = \frac{4}{9}.$$

MCQ — Evaluating limits by factoring (the 0/0 form)

43 Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$.

a. 16

b. 0

c. 8 ✓

d. 4

$$\frac{(x - 4)(x + 4)}{x - 4} \rightarrow x + 4 = 8.$$

44 Evaluate $\lim_{x \rightarrow -3} \frac{x^2 + 7x + 12}{x + 3}$.

a. 7

b. 4

c. -1

d. 1 ✓

$$\frac{(x + 3)(x + 4)}{x + 3} \rightarrow x + 4 = 1.$$

45 Evaluate $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$.

a. 0

b. 5

c. 10 ✓

d. 25

$$\frac{(x - 5)(x + 5)}{x - 5} \rightarrow x + 5 = 10.$$

46 Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 5x + 6}$.

a. -1

b. $\frac{4}{3}$

c. -4 ✓

d. 4

$$\frac{(x-2)(x+2)}{(x-2)(x-3)} \rightarrow \frac{x+2}{x-3} = \frac{4}{-1} = -4.$$

47 Evaluate $\lim_{x \rightarrow -1} \frac{x^2 - 1}{x^2 + 3x + 2}$.

a. -2 ✓

b. 2

c. -1

d. $-\frac{1}{2}$

$$\frac{(x-1)(x+1)}{(x+1)(x+2)} \rightarrow \frac{x-1}{x+2} = \frac{-2}{1} = -2.$$

MCQ — Limits at infinity

48 Evaluate $\lim_{x \rightarrow \infty} \frac{4x^2 + 3x}{2x^2 - 7}$.

a. 2 ✓

b. 0

c. $\frac{3}{7}$

d. 4

Same degree: ratio of leading coefficients $= \frac{4}{2} = 2$.

49 Evaluate $\lim_{x \rightarrow \infty} \frac{3x + 5}{x^2 + 2}$.

a. 3

b. $\frac{5}{2}$

c. ∞

d. 0 ✓

The denominator's degree is higher: the limit is 0.

50 Evaluate $\lim_{x \rightarrow \infty} \frac{7x^3 - 1}{x^3 + 3x}$.

a. 0

b. 1

c. 7 ✓

d. -1

Same degree: ratio of leading coefficients $= \frac{7}{1} = 7$.

51 Evaluate $\lim_{x \rightarrow \infty} \frac{2x^2 + 1}{5x^3 - 4}$.

a. ∞

b. $\frac{2}{5}$

c. 0 ✓

d. 5

The denominator's degree is higher: the limit is 0.

MCQ — One-sided limits

52 A function is defined by $f(x) = x + 2$ for $x < 3$, and $f(x) = 3x - 4$ for $x \geq 3$. Find $\lim_{x \rightarrow 3^-} f(x)$.

a. 3

b. 4

c. 9

d. 5 ✓

Using $x < 3$: $f(x) \rightarrow 3 + 2 = 5$.

53 Using the same piecewise function, find $\lim_{x \rightarrow 3^+} f(x)$.

a. 3

b. 4

c. 5 ✓

d. 9

Using $x \geq 3$: $f(x) \rightarrow 3(3) - 4 = 5$.

54 Using the same piecewise function, does $\lim_{x \rightarrow 3} f(x)$ exist? State its value.

a. Yes, it equals 9

b. No, it does not exist

c. Yes, it equals 5 ✓

d. Yes, it equals 4

Since the left and right limits both equal 5, the two-sided limit exists and equals 5.

55 A function is defined by $f(x) = 2x$ for $x \leq 1$, and $f(x) = x + 3$ for $x > 1$. Find $\lim_{x \rightarrow 1^-} f(x)$.

a. 1

b. 2 ✓

c. 3

d. 4

Using $x \leq 1$: $f(x) \rightarrow 2(1) = 2$.

56 Using the same piecewise function, find $\lim_{x \rightarrow 1^+} f(x)$, and state whether $\lim_{x \rightarrow 1} f(x)$ exists.

a. 4; does not exist ✓

b. 4; exists

c. 2; exists

d. 2; does not exist

Right limit = $1 + 3 = 4$. Since the left limit (2) differs from the right limit (4), the two-sided limit does not exist.

MCQ — The power rule

57 Differentiate $f(x) = x^7$.

a. x^6

b. $7x^8$

c. $6x^7$

d. $7x^6$ ✓

Power rule: $7x^{7-1} = 7x^6$.

58 Differentiate $f(x) = 4x^5 - 3x^3 + 2x$.

a. $4x^4 - 3x^2 + 2$

b. $20x^4 - 3x^2 + 2$

c. $20x^4 - 9x^2$

d. $20x^4 - 9x^2 + 2$ ✓

Differentiate each term: $20x^4 - 9x^2 + 2$.

59 Differentiate $f(x) = \sqrt{x}$.

a. $\frac{1}{2}\sqrt{x}$

b. $\frac{1}{\sqrt{x}}$

c. $2\sqrt{x}$

d. $\frac{1}{2\sqrt{x}}$ ✓

$$f(x) = x^{1/2} \Rightarrow f'(x) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}.$$

60 Differentiate $f(x) = \frac{1}{x^3}$.

a. $-\frac{3}{x^2}$

b. $-\frac{3}{x^4}$ ✓

c. $\frac{3}{x^4}$

d. $-\frac{1}{3x^4}$

$$f(x) = x^{-3} \Rightarrow f'(x) = -3x^{-4} = -\frac{3}{x^4}.$$

61 Differentiate $f(x) = 6x^2 - 4x + 9$.

a. $6x - 4$

b. $12x + 9$

c. $12x - 4x$

d. $12x - 4$ ✓

Differentiate each term: $12x - 4$.

MCQ — The product rule

62 Differentiate $f(x) = x^3(2x + 5)$ using the product rule.

a. $8x^3 + 15x^2$ ✓

b. $6x^2 + 5$

c. $8x^3 + 5x^2$

d. $2x^3 + 15x^2$

$$f'(x) = 3x^2(2x + 5) + x^3(2) = 6x^3 + 15x^2 + 2x^3 = 8x^3 + 15x^2.$$

63 Differentiate $f(x) = (x + 3)(x - 7)$ using the product rule.

a. $x - 4$

b. $2x - 4$ ✓

c. $2x - 10$

d. $2x + 4$

$$f'(x) = (x - 7) + (x + 3) = 2x - 4.$$

64 Differentiate $f(x) = (3x - 2)(x^2 + 4)$ using the product rule.

a. $6x^2 - 4x + 12$

b. $3x^2 - 4x + 12$

c. $9x^2 + 12$

d. $9x^2 - 4x + 12$ ✓

$$f'(x) = 3(x^2 + 4) + (3x - 2)(2x) = 3x^2 + 12 + 6x^2 - 4x = 9x^2 - 4x + 12.$$

65 Differentiate $f(x) = (x^2 + 1)(x^2 - 1)$ using the product rule.

a. $4x^3$ ✓

b. $4x$

c. $2x^3$

d. $4x^3 - 2x$

$$f'(x) = 2x(x^2 - 1) + (x^2 + 1)(2x) = 2x^3 - 2x + 2x^3 + 2x = 4x^3.$$

66 Differentiate $f(x) = x^2(x^3 - 4)$ using the product rule.

- a. $5x^4 - 4$
- b. $3x^4 - 8x$
- c. $5x^4 - 8x$ ✓
- d. $5x^4 - 8$

$$f'(x) = 2x(x^3 - 4) + x^2(3x^2) = 2x^4 - 8x + 3x^4 = 5x^4 - 8x.$$

MCQ — The chain rule

67 Differentiate $f(x) = (2x + 3)^5$.

- a. $5(2x + 3)^4$
- b. $2(2x + 3)^4$
- c. $10(2x + 3)^5$
- d. $10(2x + 3)^4$ ✓

$$f'(x) = 5(2x + 3)^4 \cdot 2 = 10(2x + 3)^4.$$

68 Differentiate $f(x) = (x^2 - 3)^4$.

- a. $8x(x^2 - 3)^3$ ✓
- b. $4(x^2 - 3)^3$
- c. $4x(x^2 - 3)^3$
- d. $8x(x^2 - 3)^4$

$$f'(x) = 4(x^2 - 3)^3 \cdot 2x = 8x(x^2 - 3)^3.$$

69 Differentiate $f(x) = \sqrt{5x + 1}$.

a. $\frac{1}{2\sqrt{5x + 1}}$

b. $\frac{5}{2\sqrt{5x + 1}}$ ✓

c. $5\sqrt{5x + 1}$

d. $\frac{5}{\sqrt{5x + 1}}$

$$f'(x) = \frac{1}{2\sqrt{5x + 1}} \cdot 5 = \frac{5}{2\sqrt{5x + 1}}.$$

70 Differentiate $f(x) = (4 - 3x)^5$.

a. $-15(4 - 3x)^4$ ✓

b. $-5(4 - 3x)^4$

c. $-3(4 - 3x)^4$

d. $15(4 - 3x)^4$

$$f'(x) = 5(4 - 3x)^4 \cdot (-3) = -15(4 - 3x)^4.$$

71 Differentiate $f(x) = (x^3 + 2)^2$.

a. $6x^2(x^3 + 2)$ ✓

b. $6x^2(x^3 + 2)^2$

c. $3x^2(x^3 + 2)$

d. $2(x^3 + 2)$

$$f'(x) = 2(x^3 + 2) \cdot 3x^2 = 6x^2(x^3 + 2).$$

MCQ — Derivatives of trigonometric and exponential functions

72 Differentiate $f(x) = \cos x - \sin x$.

a. $\sin x + \cos x$

b. $-\sin x - \cos x$ ✓

c. $\sin x - \cos x$

d. $-\sin x + \cos x$

$$\frac{d}{dx}[\cos x] = -\sin x, \frac{d}{dx}[-\sin x] = -\cos x.$$

73 Differentiate $f(x) = 5\cos x$.

a. $5\sin x$

b. $-5\cos x$

c. $5\cos x$

d. $-5\sin x$ ✓

$$\frac{d}{dx}[5\cos x] = -5\sin x.$$

74 Differentiate $f(x) = e^x - x^3$.

a. $e^x - 3x$

b. $e^x - x^2$

c. $xe^{x-1} - 3x^2$

d. $e^x - 3x^2$ ✓

$$\frac{d}{dx}[e^x] = e^x, \frac{d}{dx}[-x^3] = -3x^2.$$

75 Differentiate $f(x) = 2e^x + \sin x$.

a. $2e^x + \cos x$ ✓

b. $2e^x + \sin x$

c. $e^x + \cos x$

d. $2e^x - \cos x$

$$\frac{d}{dx}[2e^x] = 2e^x, \frac{d}{dx}[\sin x] = \cos x.$$

MCQ — Tangent lines

76 Find the slope of the tangent line to $f(x) = x^2$ at $x = 4$.

- a. 4
- b. 2
- c. 16
- d. 8 ✓

$$f'(x) = 2x; f'(4) = 8.$$

77 Find the equation of the tangent line to $f(x) = x^2$ at $x = 3$.

- a. $y = 9x - 6$
- b. $y = 6x + 9$
- c. $y = 3x - 9$
- d. $y = 6x - 9$ ✓

$$f(3) = 9, f'(3) = 6: y - 9 = 6(x - 3) \Rightarrow y = 6x - 9.$$

78 The graph of $f(x) = x^2 - 2x$ is shown, with the point at $x = 3$ marked. Find the equation of the tangent line there.

- a. $y = 3x - 9$
- b. $y = 4x - 3$
- c. $y = 4x + 9$
- d. $y = 4x - 9$ ✓

$$f(3) = 3, f'(3) = 4: y - 3 = 4(x - 3) \Rightarrow y = 4x - 9.$$

79 Find the x -coordinate(s) where the tangent to $f(x) = x^3 - 12x$ is horizontal.

- a. $x = 0$
- b. $x = 4$ and $x = -4$
- c. $x = 2$ only
- d. $x = 2$ and $x = -2$ ✓

$$f'(x) = 3x^2 - 12 = 0 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2.$$

80 The graph shows $f(x) = x^2$ and its tangent line at $x = 2$. Find the slope of the tangent line by reading the graph.

- a. 4 ✓
- b. 2
- c. 1
- d. 8

The tangent line rises 4 for every 1 across.

MCQ — Increasing, decreasing, and the second derivative

81 For $f(x) = x^2 - 6x + 3$, find $f'(x)$ and determine the interval where f is increasing.

- a. $f'(x) = 2x - 6$; increasing for $x < 3$
- b. $f'(x) = 2x - 6$; increasing for $x > 3$ ✓
- c. $f'(x) = 2x$; increasing for $x > 3$
- d. $f'(x) = 2x - 6$; increasing for $x > 6$

$$f'(x) = 2x - 6 > 0 \Rightarrow x > 3.$$

82 For $f(x) = -x^2 + 8x$, determine the interval where f is decreasing.

a. $x < 4$

b. $x > 8$

c. $x > 4$ ✓

d. $x < 8$

$$f'(x) = -2x + 8 < 0 \Rightarrow x > 4.$$

83 Find the second derivative of $f(x) = x^5 - 3x^3$.

a. $20x^2 - 18x$

b. $20x^3 - 18x$ ✓

c. $20x^3 - 9x$

d. $5x^4 - 9x^2$

$$f'(x) = 5x^4 - 9x^2; f''(x) = 20x^3 - 18x.$$

84 Find the second derivative of $f(x) = x^4 + 2x^2$.

a. $12x^2$

b. $4x^3 + 4x$

c. $12x^2 + 4$ ✓

d. $12x^2 + 4x$

$$f'(x) = 4x^3 + 4x; f''(x) = 12x^2 + 4.$$

MCQ — Word problems: rates of change

- 85 The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the ball's velocity at $t = 2$ seconds.

a. 10 ✓

b. -10

c. 30

d. 20

$$h'(t) = -10t + 30; h'(2) = 10 \text{ m/s.}$$

- 86 A company's cost to produce x items is $C(x) = 0.02x^2 + 8x + 150$ (in dollars). Find the marginal cost at $x = 100$.

a. 8 dollars

b. 4 dollars

c. 20 dollars

d. 12 dollars ✓

$$C'(x) = 0.04x + 8; C'(100) = 12.$$

- 87 The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the time at which the ball reaches its maximum height.

a. $t = 3$ ✓

b. $t = 5$

c. $t = 6$

d. $t = 2$

$$h'(t) = -10t + 30 = 0 \Rightarrow t = 3 \text{ seconds.}$$